Diseases of the Arterial System

NURA 340 Pathophysiology | Lecture Study Guide

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Lecture roadmap: Arterial wall and endothelial dysfunction → Hypertension → Atherosclerosis → Peripheral artery disease → Arterial thrombosis and embolism → Aneurysms

Please note: The transcript you provided ends partway through hypertension signs and symptoms, so everything after that point comes from the slides only. Those sections are marked "From the slides" and contain no lecture commentary.

The Arterial Wall and Endothelial Dysfunction

Under normal conditions, arteries act as high-pressure, high-elastic conduit and resistance vessels. Each has three layers:

- **Tunica intima:** the inner layer, a single layer of endothelial cells lining every blood vessel in the body.
- **Tunica media:** the smooth muscle layer that controls vessel diameter and resistance.
- **Tunica adventitia:** the outer protective layer.

The Healthy Endothelium

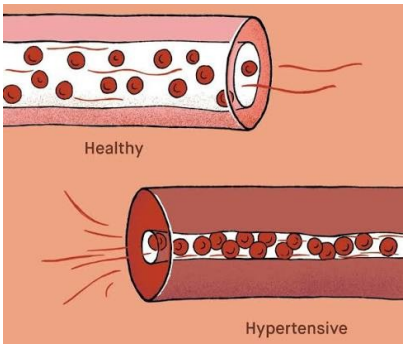
At the center of normal arterial function is the endothelium, which acts as an active endocrine organ. It continuously synthesizes and releases nitric oxide to maintain baseline vasodilation, inhibits smooth muscle proliferation, prevents platelet aggregation, and regulates vascular permeability (think inflammation).

Endothelial Injury

Arterial disease almost always begins with endothelial injury and dysfunction. Damage may come from high mechanical wall tension, toxic lipid molecules, or systemic chemical exposures. The damaged endothelium:

- loses its ability to produce nitric oxide,
- becomes hyperpermeable, and
- becomes sticky to inflammatory cells.

Key concept: This mechanism drives nearly every condition in this lecture. Keep it in mind throughout.



Healthy vs. hypertensive artery (lecture slide)

Hypertension

Priority topic: Hypertension is the condition to know best. It is the most common arterial disease, and every kind of nurse constantly educates about it, treats it, and tries to prevent it.

Definition: systolic blood pressure above 130 mmHg and/or diastolic blood pressure above 80 mmHg. It is classified into stages of disease progression.

What Blood Pressure Measures

Arterial blood pressure is the lateral force exerted by circulating blood against the walls of the arteries.

- **Systolic (SBP):** the peak arterial pressure during ventricular contraction. It reflects stroke volume, velocity of ejection, and the compliance (stretchability) of the aorta and large arteries.
- **Diastolic (DBP):** the minimum pressure maintained during relaxation (diastole). It is governed mainly by systemic vascular resistance and the elastic recoil of large arteries.
- **Mean arterial pressure (MAP):** the average pressure pushing blood through the capillary beds over the whole cardiac cycle. Diastole lasts about twice as long as systole at a normal heart rate, so:

$$MAP = DBP + \tfrac{1}{3}(SBP - DBP)$$

You will not be asked to calculate MAP on the test, but you should understand how it is derived. The baseline for a healthy adult is a MAP of 65 mmHg, the minimum required to maintain organ perfusion. Below this threshold, tissue beds can become ischemic.

Mechanism: Blood pressure is controlled by cardiac output and systemic vascular resistance. Hypertension occurs when there is an uncompensated, sustained increase in cardiac output, systemic vascular resistance, or both.

Classification

	Primary hypertension	Secondary hypertension
Frequency	90–95% of cases	5–10% of cases
Cause	No single discrete cause; multifactorial, from genetic predisposition interacting with environmental stressors	An underlying identifiable condition or medication, most often renal or endocrine dysfunction
Mechanism	Multi-system regulatory breakdown; normal negative feedback loops fail to lower pressure back to baseline	Depends on the cause; treating the root cause often resolves or significantly improves the blood pressure

Primary Hypertension

Risk factors

- **Genetic / non-modifiable:** family history of hypertension; Black, Asian, or Hispanic race/ethnicity; being male under 65 or female over 65 (age- and sex-related).
- **Modifiable:** low physical activity, smoking, alcohol use, stress, obesity, obstructive sleep apnea, and poor dietary choices.

Pathophysiology

Arterial pressure is regulated by high-pressure and low-pressure baroreceptors that signal the central nervous system, adjusting tension and resistance of the peripheral vasculature. In response to low pressure they trigger release of antidiuretic hormone (ADH), renin, and aldosterone (the RAS), which affect blood volume and vascular resistance.

With the risk factors above, one or both of these feedback loops becomes desensitized to high blood pressure over time. Because the receptors are used to the elevated pressure, they stop sending the corrective signals and the response weakens as pressure rises.

Secondary Hypertension

Causes / risk factors

- Renal disease
- Congenital vascular disorders (present from birth)
- Obstructive sleep apnea
- Endocrine disorders such as Cushing syndrome and polycystic ovarian syndrome (Cushing's is covered in the endocrine unit)
- Stimulant use, caffeine, and excessive alcohol
- Pregnancy
- Adverse reaction to medications used to treat other disorders; sometimes medications must be changed because they are causing hypertension

Pathophysiology

- **Renal dysfunction:** (1) disrupts fluid balance, increasing intravascular volume and blood pressure; (2) can trigger the RAS, raising pressure through aldosterone release (too much, or released when it is not needed).
- **Endocrine dysfunction:** organ dysfunction leads to excess aldosterone secretion from the kidneys or adrenal glands, altering intravascular volume.

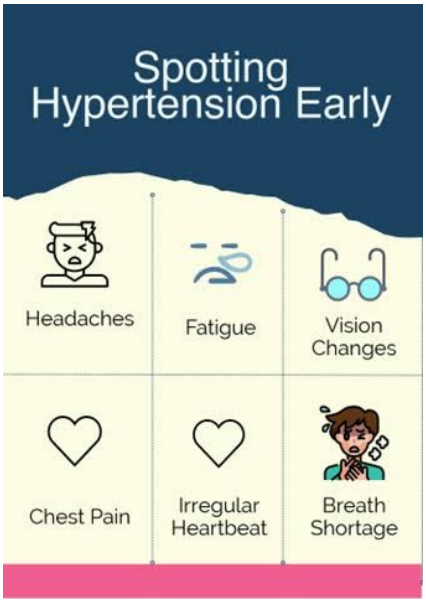
The pathways are similar; one is triggered by renal dysfunction and the other by endocrine dysfunction. Control through the sympathetic nervous system, RAS pathways, and kidney pressure regulation becomes chronically dysregulated.

Signs and Symptoms

Blood pressure regulation is a chemical and physical response within the arterial system, so many patients have no manifestations at all and cannot feel that they are hypertensive. This is why there is a push for preventative monitoring, such as free blood pressure machines at pharmacies and fitness clubs, to catch hypertension early, before symptoms and permanent damage develop.

Some patients have non-specific symptoms (headache, dizziness, shortness of breath). Additional signs of kidney or endocrine dysfunction help providers decide whether the hypertension is secondary. The symptoms of hypertension itself are the same whether primary or secondary:

- **Headache:** high pressure overwhelms cerebral autoregulation, causing hyperperfusion, blood-brain barrier breakdown, and eventually brain swelling in hypertensive crisis.
- **Chest pain and shortness of breath:** the heart works harder against high resistance, raising myocardial oxygen demand. If coronary flow cannot keep up, ischemia results. Rising left-sided pressure can back fluid up into the lungs.
- **Vision changes:** high pressure damages the small retinal vessels.
- **Fatigue:** from increased metabolic demand.



Spotting hypertension early (lecture slide)

Hypertensive Crisis: Urgent or Emergent?

From the slides

Feature	Hypertensive urgency	Hypertensive emergency
BP threshold	Typically > 180/120 mmHg	Typically > 180/120 mmHg
Acute target organ damage?	NO active, evolving organ damage	YES: active, evolving organ dysfunction
Pathophysiology	Severe elevation in SVR, but autoregulation in brain, heart, and kidneys remains intact	Extreme pressure overcomes vascular autoregulation, causing endothelial necrosis, acute exudation, and tissue edema
Presentation	Asymptomatic, or mild headache, severe anxiety, nosebleeds	Encephalopathy, intracranial hemorrhage, MI, pulmonary edema, renal failure

Hypertension Diagnostics

From the slides

- **Testing:** blood pressure monitoring, EKG, echocardiogram, renal ultrasound.
- **Kidney function:** creatinine, BUN, eGFR.
- **Lipid panel**
- **Blood glucose / A1C**
- **Electrolytes:** sodium, potassium, calcium.

Atherosclerosis

From the slides

A chronic, progressive inflammatory disease characterized by lipid accumulation and fibrous plaque formation.

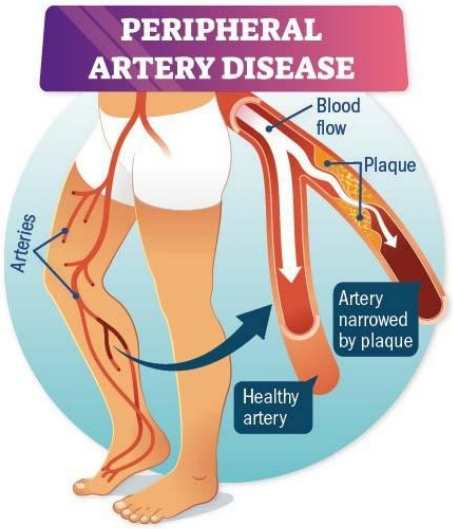
	Details
Non-modifiable risks	Age, genetics, male gender
Modifiable risks	Hyperlipidemia, hypertension, smoking, diabetes mellitus, obesity
Signs / symptoms	Asymptomatic; pain
Diagnostics	Lipid panel; CRP; blood glucose and A1C; coronary artery calcium scoring

Peripheral Artery Disease (PAD)

From the slides

Atherosclerotic plaques form in systemic arteries outside the heart and brain, most commonly in the lower extremities.

- **Risk factors:** smoking, diabetes mellitus, advanced age, hyperlipidemia, hypertension.
- **Signs / symptoms:** intermittent claudication; diminished peripheral pulses; pale, shiny skin; hair loss; cool extremities; pain; ischemic ulcers.
- **Diagnostics:** pulse checks, ultrasound, CT angiography or MRA.



Peripheral artery disease (lecture slide)

Arterial Thrombosis and Embolism

From the slides

	Arterial thrombosis	Arterial embolism
Definition	Blood clot forming over an atherosclerotic plaque	Mobile clot that lodges in a small arterial vessel
Risk factors	Atherosclerosis, vascular trauma, hypercoagulable states	Atrial fibrillation, MI, prosthetic heart valves, infective endocarditis

Signs and Symptoms: The 6 P's

Pain, Pallor, Pulselessness, Poikilothermia (temperature change), Paresthesia, and Paralysis.

Diagnostics

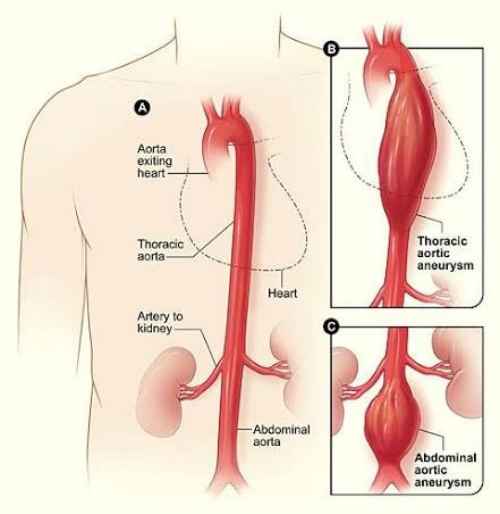
- **Imaging:** bedside Doppler, digital subtraction angiography, catheter angiography.
- **Labs:** creatine kinase and myoglobin; coagulation studies (PT/INR, aPTT, fibrinogen).

Arterial Aneurysms

From the slides

The arterial wall weakens and progressively bulges outward.

- **Risk factors:** atherosclerosis, hypertension, connective tissue disorders, advanced age, male gender, smoking.

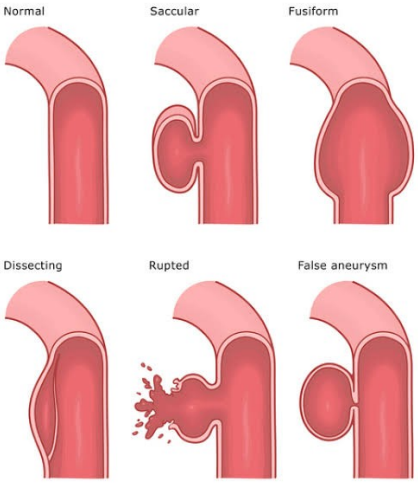


Aortic aneurysm locations (lecture slide)

Abdominal vs. Thoracic Aortic Aneurysm

Feature	Abdominal (AAA)	Thoracic (TAA)
Dominant pathology	Advanced atherosclerosis, hypertension, age	Cystic medial necrosis, connective tissue disorders
Local compression signs	Lower back / flank ache, early satiety, epigastric discomfort	Hoarseness, dysphagia, superior vena cava syndrome
Key signs / symptoms	Palpable, pulsatile abdominal mass; vascular bruit	Non-palpable; possible unequal upper extremity BP or pulses
Primary diagnostic	Abdominal ultrasound	Chest CT or transesophageal echo

- **Labs for rupture / dissection:** CBC and H/H, D-dimer, type and screen.



Types of aneurysm (lecture slide)

Notes on Transcript Cleanup

Corrected mis-transcriptions: "epithelial" cells/injury is endothelial (the lecture and slides refer to the endothelium); "tunica adventia" is adventitia; "platelet aggravation" is aggregation; "intervascular" is intravascular; "milligrams of mercury" is mmHg. One sentence says blood pressure control is dysregulated "for hypotensive patients"; I read this as hypertensive, which fits the context. The transcript ends mid-sentence at fatigue, and the slides on hypertensive crisis onward had no matching transcript.